

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Odd Semester Mid-term Examination, 2025-26

Course Code: ESC331

Full Marks: 25

Course Name: Geology for Civil Engineering

Time: 90 Minutes

Question Paper No.: NITDGP/ESC331/MID01

Instructions: Answer any five answers.

Question No.	Body of the Question	Marks
1	Write three reasons about the importance of geology in civil engineering.	5
2	(a) Mention the types of minerals on the basis of colour and also give example of one mineral from each type. (b) Define lustre of a mineral. (c) What type of lustre quartz exhibits? <i>→ glassy vitreous</i>	3 + 1 + 1 = 5
3	(a) Define hardness of a mineral. (b) 'Graphite and diamond exhibit different degree of hardness although chemically they have same composition' – give reason. (c) Differentiate haematite from magnetite using streak.	1 + 2 + 2 = 5
4	(a) Describe the texture of igneous rock based on crystallinity. (b) 'Volcanic rocks are fine grained rock whereas plutonic rocks are coarse grained rock' – give reason.	3 + 2 = 5
5	(a) Write the names of the factors that are used to determine sedimentary texture. (b) What does 'degree of roundness' of grains indicates? (c) Describe graded bedding in sedimentary rock. Also draw a sketch diagram.	2 + 1 + 2 = 5
6	(a) Write the role of temperature in metamorphism. Also name a type of metamorphism in which temperature plays an important role. <i>→ Regional</i> (b) Which metamorphism is responsible for transformation of limestone to marble? <i>→ Contact</i> (c) Define metamorphic grade. Also mention their types.	3 + 1 + 1 = 5
7	(a) Write a brief note on zones of subsurface water. Also draw a sketch. (b) Define confined aquifer. (c) Define hydraulic gradient. <i>→ ratio of drop in head to distance</i> (d) The water level in a well decreases by 12 metres over a horizontal distance of 120 metres to another well. Determine the hydraulic gradient. <i>120</i>	2 + 1 + 1 + 1 = 5
8	(a) Define electrical resistivity of a rock. (b) Discuss the electrical resistivity method for ground water exploration.	1 + 4 = 5